



95-Which of the following statements concerning the partitioning of the atria is CORRECT?

- a) The septum secundum is a transient structure which degenerates by the end of the 5th week
b) The ostium secundum forms by apoptosis in the central part of the septum
c) The septum primum forms the valve of the oval foramen
d) The ostium primum is a common atrial-septal defect

96-A newborn infant presents with cyanotic heart disease and fails to thrive. On examination of the heart, it was revealed that the aorta arose from the right ventricle. The pulmonary trunk arose from the left ventricle. What is the name given to this cardiac malformation?

- a) Persistent truncus arteriosus
b) Transposition of the great vessels
c) Tetralogy of Fallot
d) Ventricular septal defect (VSD)

97-The sinus venarum is

- a) Formed from the right auricle.
b) The trabeculated part of the right atrium.
c) Formed from tissue of the right sinus horn
d) Formed from the pulmonary veins.

98-The internal carotid artery is derived from

- a) the distal part of the third arch artery and cranial most part of the dorsal aorta
b) Third aortic arch
c) Dorsal aorta
d) Second and third aortic archs

99-The SVC is derived from:

- a) Right cardinal vein
b) Right posterior cardinal vein
c) Right cardinal and right common cardinal
d) Right common cardinal

Continue to next page

88- Which one has dual embryonic origin ?

- a) lymphnode
- b) spleen
- c) tonsil
- d) thymus

89- We observe this type of capillaries in bone marrow and spleen ?

- a) continuous
- b) fenestrated with diaphragm
- c) fenestrated without diaphragm
- d) sinusoids

90- Which one is not monomer ?

- a) IgA
- b) IgE
- c) IgD
- d) IgG

91- Which one is partially capsulated ?

- a) thymus
- b) spleen
- c) tonsil
- d) lymphnode

92- Where is the site of destruction of aged erythrocytes ?

- a) spleen
- b) thymus
- c) lymphnode
- d) tonsil

Embryology: Dr. Amidi

93- Which part of the primitive heart tube gives rise to the pulmonary artery ?

- a) Bulbus cordis
- b) Primitive atrium
- c) Primitive ventricle
- d) Truncus arteriosus

94- The left sinus horn becomes the

- a) Coronary sinus.
- b) Inferior vena cava
- c) Oblique vein of the right atrium
- d) Main inflow of the right atrium



83- In the radiology image, the shadow on the right side of the heart belongs to cavity

- a) Right atrium
- b) Left atrium
- c) Right ventricle
- d) Left ventricle

84- Pectineal line belong to which of the following bone?

- a) Sacrum
- b) Coccyx
- c) Pubis
- d) Ilium

85- All the following muscles are attached to the spinous process except...?

- a) Trapezius
- b) Serratus posterior inferior
- c) serratus anterior
- d) serratus posterior superior

Histology: Dr. Malek

86- Most numerous vessels in our body are ...?

- a) small arteries
- b) veins
- c) capillaries
- d) large arteries

87- The thickest layer in the wall of arteries are ?

- a) adventitia
- b) intima
- c) media
- d) serosa



77- All of the following are involved in the formation of the lower thoracic opening

- a) 11 vertebrae
- b) 12 ribs
- c) 11 ribs
- d) Costal margin

78- Which of the following muscles are innervated by the posterior branch of the nerves?

- a) Longissimus thoracic
- b) Serratus posterior inferior
- c) Sub costal
- d) Rhomboid minor

79- Intercostal nerve innervates all areas below except...

- a) External intercostal muscles
- b) Visceral pericardium
- c) Transverse thoracic
- d) Intima intercostal muscles

80- The Diagonal artery branches from which of the following arteries?

- a) Anterior interventricular
- b) Circumflex
- c) Marginal
- d) Right coronary

81- The descending aorta posteriorly is adjacent to which of the following

- a) Hemiazygos vein
- b) Esophagus
- c) Pericard
- d) Left ventricle

82- Which of the following statement is not correct?

- a) Phrenic nerve innervate fibrous pericard
- b) Thoracic duct is passing from anterior mediastinum
- c) Left upper limb lymph drain into thoracic duct
- d) Parasympathetic nerve in thorax coming from vagus nerve



66- The ascending aorta is posteriorly adjacent to which of the following elements?

- a) Vena cava Superior
- b) Left ventricle
- c) Coronary sinus
- d) Right pulmonary artery

67- Which of the following structures is adjacent to the posterior and right aortic arch?

- a) Right phrenic
- b) left vagus
- c) Left recurrent laryngeal nerve
- d) Ligamentum arteriosum

68- Which of the following sentences is correct about the sympathetic chain?

- a) Nerve fibers in the greater splanchnic nerve have myelination
- b) The fourth sympathetic ganglia is involved in the formation of the splanchnic nerve
- c) White communication rami is visible in cervical portion of sympathetic chain.
- d) Impar ganglion is visible in thoracic portion of sympathetic chain.

69- Which of the cardiac cavities is in the posterior position compared to the others in the chest?

- a) Right atrium
- b) Left atrium
- c) Left ventricle
- d) right ventricle

70- The transverse pericardial sinus is anteriorly adjacent to which of the following elements?

- a) Pulmonary artery trunk
- b) Vena cava inferior
- c) Descending aorta
- d) Vena cava Superior



71- All the following structures are seen in the superior mediastinum except

- a) Left phrenic ✓
- b) Thymus ✓
- c) Right phrenic ✓
- d) Ascending aorta ✓

72- Where is the auscultation of the pulmonary valve in the surface anatomy?

- a) apex of the heart
- b) The second right intercostal space
- c) the fifth left intercostal space
- d) The second left intercostal space

73- Which of the following muscles are active in elevation of the ribs?

- a) Serratus Posterior Inferior
- b) Transverse thoracis
- c) Subcostal
- d) External intercostal

74- All the following veins drain into the right atrium except

- a) Coronary sinus
- b) Vena cava inferior
- c) Right pulmonary veins
- d) Vena Cava Superior

75- Which of the following areas can be seen in the anterior wall of the pericardial sinus?

- a) Left atrium
- b) Right atrium
- c) Esophagus
- d) Right pulmonary veins

76- Which of the thoracic vertebrae are posterior of the base of the heart position

- a) 3 to 7
- b) 5 to 8

- c) 6 to 9
- d) 7 to 10



Anatomy: Dr. Pasbakhsh

61- Which of the following ligaments is connected to the non-articular part of the rib?

- a) Posterior longitudinal .Lig
- b) Interspinous.Lig
- c) Lateral costotransverse.Lig
- d) Flavoum.Lig

62- All the following related to the left ventricle except...?

- a) Papillary muscles
- b) Chorda tendino
- c) Apex
- d) Septo marginal

63- Which of the following joint is fibrocartilage type ?

- a) Third costovertebral
- b) Third costotransverse
- c) First costo sternal
- d) Zygapophysial

64- The ninth posterior intercostal vein on the left side drains into which of the veins?

- a) Azygos
- b) Accessory hemizygous
- c) Hemizygous
- d) Internal thoracic

65- The ninth anterior intercostal artery is branches from which of the following?

- a) Muscloprenic
- b) Internal thoracic
- c) Superior epigastric
- d) Ascending aorta



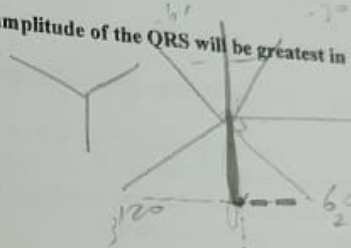
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56-Which of the following leads has an electrical axis of -30 degrees

- a) Lead I
- b) Lead III
- c) Lead aVL
- d) Lead aVF

57-If the mean electrical axis is 90° , then the net amplitude of the QRS will be greatest in

- a) Lead I
- b) Lead III
- c) aVF
- d) aVL



58-The last portion of the heart to depolarize is

- a) the entire outer surface of the apex
- b) the endocardial areas
- c) the inner surface of the apex
- d) posterobasal portion of the left ventricle and the pulmonary conus

59-In a Lead I ECG recording, the first small negative deflection of the QRS (the Q-wave) represents

- a) Atrial repolarization
- b) Depolarization of the interventricular septum from left to right
- c) A wave of depolarization traveling towards the left leg
- d) Ventricular repolarization

60-Leads V5-V6 overlie which region of the left ventricle

- a) Anteroapical
- b) Anterolateral
- c) Anteroseptal
- d) Inferior



Physiology: Dr. Ashabi, Dr. sohanaki, Dr. Rkhshan

1- The ... have the largest total cross-sectional area of the circulation.

- a) capillaries ✓
b) aorta
c) veins
d) arteries

2- Which vessel has greatest blood velocity?

- a) Aorta ✓
b) Vena cava
c) Arterioles X
d) Capillaries X

3- In which one, huge volume of blood could has little effect on pressure

- a) Aorta
b) Bronchi
c) Arteries
d) Veins ✓

4- How would a decrease in vascular resistance affect blood flow?

- a) Reduce blood flow
b) Has no role
c) May be reduce blood flow
d) Increase blood flow ✓

5- ... is the quantity of blood that passes a given point in the circulation time.

- a) Resistance
b) Blood pressure
c) Blood flow ✓
d) Capacitance

6- Laminar flow is ..., whereas turbulent flow tend to cause murmurs.

- a) happened by high velocity
- b) reverse blood flow
- c) silent
- d) has no parabolic profile

7- Based on Poiseuille's law, which factor has more effect on blood flow?

- a) Radius of vessel
- b) Viscosity of blood
- c) Length of vessel
- d) Pressure of blood

8- In auscultatory method for measuring blood pressure, which sound systolic pressure sound?

- a) First
- b) last
- c) none
- d) third

9- Pressure in the ... is called central venous pressure.

- a) aorta
- b) capillaries
- c) right atrium
- d) large arteries

10- Which factor could increase venous return?

- a) Reduced blood volume
- b) Reduced large vessel tone resulting
- c) Reduced peripheral venous pressure
- d) Dilation of the arterioles



11-Valves are arranged so that so that the direction of blood flow can

- a) legs
- ☒ b) Heart
- c) hands
- d) feet

12-Which one could increase cardiac output?

- ☒ a) Anemia
- b) Cardiac shock
- c) Traumatic shock
- d) Heart valves disease

13-Renin-angiotensin II-aldosterone is responding to

- a) Arterial pressure reduction
- b) Veins pressure reduction
- ☒ c) Arterial pressure enhancement
- d) Veins pressure enhancement

14-Innervation of small arteries and arterioles allow sympathetic

- a) reduce viscosity
- b) increase viscosity
- c) increase vascular resistance
- ☒ d) reduce vascular resistance

15-In vessels, the neurotransmitter for the vasoconstrictor n

- ☒ a) Acetylcholine
- b) Histamin

- c) Nor
- d) En

16-Which of following is NOT correct about platelets?

- ☒ a) They have no ability for ATP production
- b) They have contraction filaments
- c) They are able to become sticky
- d) They are able to form pseudopods

17-In 80-year-old person, which part of body produce RBC?

- a) Lymph nodes
- b) Femur
- c) Liver
- d) ribs

18-All of followings are same at both neutrophil and macrophage, except

- a) Amount of phagocytized particles
- b) formation of pseudopodia
- c) have proteolytic enzym
- d) have chemotaxis

19-Inflammation is characterized by all of following signs, except

- a) swelling
- b) heat
- c) decrease capillary permeability
- d) redness

20-What is the relationship between hypoxia, EPO and RBC?

- a) Hypoxia decreases gene expression of EPO and finally RBC production
- b) Hypoxia decreases gene expression of EPO and finally RBC production
- c) Hypoxia increases gene expression of EPO and finally RBC production
- d) Hypoxia increases gene expression of EPO and finally RBC production

21-Vitamin B12.....

- a) Produced by normal flora at large intestine is absorbed at small intestine
- b) Provided by diet is absorbed at first segment of small intestine
- c) is protected from gastric digestion by attachment to R-BINDER
- d) don't store at body cells

22-Which of following acts as anti-coagulant factor ?

- a) Protein C
- b) Factor XII active
- c) Factor X active
- d) Factor VIII active



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23- Production site of neutrophil is

- ☐ a) Only lymphatic nodes
☐ b) Liver
☒ c) Bone marrow
☐ d) Only spleen

24- Which of the following cell exist at bloodstream?

- ☐ a) Proerythroblast
☐ b) Orthochromatic erythroblast
☒ c) Reticulocyte
☐ d) Basophil erythroblast
- Leukemia
Granulocytes
spleen?*

25- Which of the following statement is true?

- ☐ a) Polycythemia means decrease in hemoglobin content. X
☐ b) Polycythemia lead to decrease in blood viscosity. X
☐ c) Polycythemia increase cardiac output sharply.
☒ d) Polycythemia may occurs at healthy persons. ✓

26- In which option, the "first line defense" of the immunity system is men

- ☐ a) Increased production of granulocyte be bone marrow
☒ b) Tissue macrophages
☐ c) Neutrophil diapedesis & chemotaxis
☐ d) Secondary invasion of macrophages
- Gr*

27- Which of has shorter life span?

- ☐ a) Mature healthy RBC
☐ b) Mature healthy monocyte
☒ c) Mature healthy neutrophil at blood
☐ d) Lymphocytes

28- What do you know about difference between RBC count among n

- ☐ a) RBC number is higher in men in comparison to women due to hi
☒ b) RBC number is higher in men in comparison to women due to m
☐ c) RBC number is higher in women in comparison to men due to n
☐ d) RBC number is higher in women in comparison to men due to l



29- Native cells in our body are protected from phagocytosis by....

- a) Their smooth surface
- b) Their protective protein coat
- c) Not binding to antibody
- d) All of them are correct

30- Which type of cell produced from megakaryocytes?

- a) Platelets
- b) RBCs
- c) neutrophils
- d) macrophages

31- Inhibition of the activity of the sarcolemmal $\text{Na}^+/\text{K}^+-\text{ATPase}$ pump

- a) decreased potassium equilibrium potential
- b) increased intracellular potassium concentration ✓
- c) increased sodium loss from the cell ✗
- d) membrane hyperpolarization ✗

32- The initial depolarization during a cardiac muscle action potential

- 1. a large decrease in intracellular potassium concentration
- 2. a large increase in intracellular sodium concentration
- 3. an increase in the relative conductance of potassium
- 4. opening of fast sodium channels

33- Phase 3 of non-pacemaker action potentials results primarily

- a) decreased activity of the $\text{Na}^+/\text{K}^+-\text{ATPase}$ pump
- b) inactivation of fast sodium channels
- c) increased calcium influx through L-type channels
- d) increased potassium conductance

34- The duration of the effective refractory period in cardiac muscle

- a) inhibiting potassium channels
- b) opening L-type calcium channels
- c) shortening phase 2
- d) shortening phase 3



35-Conduction velocity in non-nodal cardiac cells is decreased by

- a) activation of fast sodium channels
- b) sympathetic nerve activation
- c) decreasing the slope of phase 0
- d) activation of SERCA 2a

36-Binding of calcium to _____ is a primary mechanism in cardiac muscle tension development.

- a) Actin
- b) Myosin
- c) troponin-C
- d) Tropomyosin

37-During the process of cardiac excitation-contraction coupling

- a) ATPase bound to actin hydrolyzes ATP to supply energy
- b) ATP is required for contraction but not relaxation
- c) calcium enters the cell through ryanodine receptors
- d) calcium enters the cell and stimulates calcium release by the sarcoplasmic reticulum

38-Calcium is transported out of the cardiac cell by

- a) passive diffusion down its concentration gradient
- b) the Ca^{++} -ATPase pump directly coupled to the $\text{Na}^{+}/\text{K}^{+}$ -ATPase
- c) the $\text{Na}^{+}/\text{Ca}^{++}$ -exchanger
- d) the $\text{Na}^{+}/\text{K}^{+}$ -ATPase pump

39-The T wave of the electrocardiogram begins to occur during

- a) early diastole
- b) isovolumetric contraction
- c) isovolumetric relaxation
- d) systolic ejection

40-The "C" wave of atrial pressure curve is due to

- a) atrial systole
- b) ventricular systole
- c) atrial diastole
- d) ventricular diastole



41-Ventricular systole is defined as the time between the

- a) closing of the aortic valve and the opening of the mitral valve
- b) closing of the mitral valve and the closing of the aortic valve
- c) opening and closing of the aortic valve
- d) opening of the aortic valve and the opening of the mitral valve

42-A hospitalized patient has an ejection fraction of 0.4, a heart rate of 100 b/min, and a cardiac output of 3.5 l/min, what is the patient's end-diastolic volume?

- a) 14,0 ml
- b) 37,0 ml
- c) 87,0 ml
- d) 92 ml

43-All cardiac valves are normally closed during which of the following p cycle?

- a) atrial contraction
- b) systolic ejection
- c) isovolumetric relaxation
- d) ventricular filling

44-The maximal rate of pressure development within the left ventricle

- a) during isovolumetric contraction
- b) during rapid ventricular ejection
- c) at peak left ventricular systolic pressure
- d) during rapid ventricular filling

45-Closure of the aortic and pulmonic valves produces which heart sound?

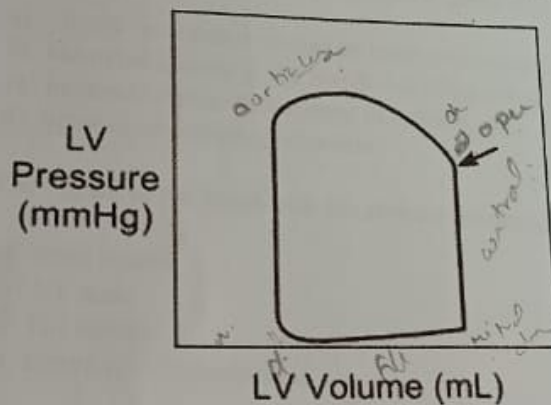
- a) S1
- b) S2
- c) S3
- d) S4



46-The mitral valve normally opens at the beginning of diastolic filling because

- a) left atrial contraction forces open the valve
- b) left ventricular pressure falls below the left atrial pressure
- c) the papillary muscles contract
- d) the papillary muscles relax

47-The following figure represents a normal left ventricular pressure-volume loop. An arrow is pointed to what event in the cardiac cycle? (Guyton p:118)



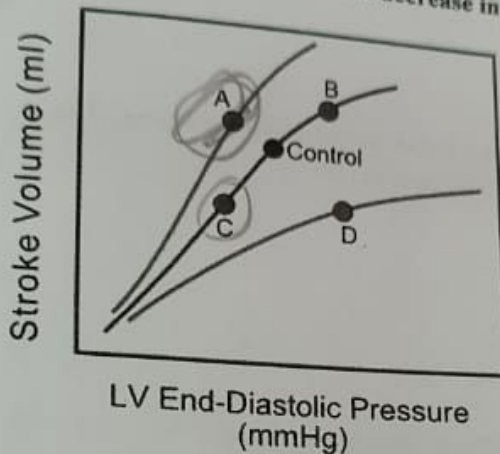
- a) aortic valve opening
- b) end-systolic volume
- c) end of systolic ejection
- d) mitral valve opening

48-Ventricular preload is increased in response to

- a) decreased atrial contractility
- b) decreased blood volume
- c) decreased ventricular inotropy
- d) increased venous compliance

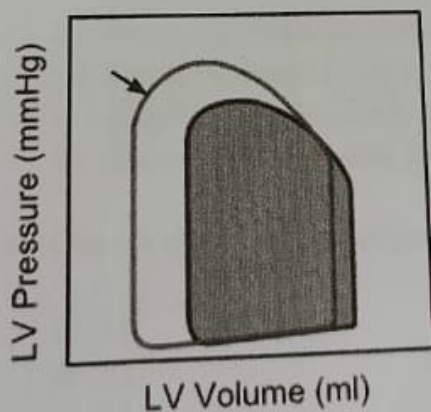


49-In the figure, left ventricular stroke volume is plotted against left ventricular end-diastolic pressure. Relative to the normal control point, which one of the following represents the independent effects of a decrease in blood volume? (Guyton p:118)



- a) point A
- b) point B
- c) point C
- d) point D

50-Compared to a normal left ventricular pressure-volume loop, the loop indicated by the arrow represents (Guyton p:118)



- a) increased preload, afterload, and inotropy
- b) increased inotropy and afterload, and decreased preload
- c) increased stroke volume with increased end-systolic volume
- d) increased stroke volume and decreased ejection fraction



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51-Ventricular muscle is isolated and set up in an apparatus to study its contractions. At a fixed preload and inotropic state, decreasing the afterload on the ventricular muscle fiber

- a) decreases the amount of fiber shortening during contraction
- b) increases active tension development by the fiber
- c) increases the rate of fiber shortening
- d) increases V_{max} when shortening velocity is plotted against afterload

52-Phase 4 of SA nodal action potentials is generated in part by

- a) "funny" currents produced by inward movement of positively charged ions
- b) increased activity of the Na^+/K^+ -ATPase pump
- c) increased outward movement of calcium ions
- d) opening of potassium channels

53-The region of the heart with the slowest conduction velocity is the

- a) atrial muscle
- b) AV node
- c) His bundle
- d) ventricular Purkinje fiber

54-The T wave of the electrocardiogram is an electrical representation of

- a) atrial depolarization
- b) atrial repolarization
- c) ventricular depolarization
- d) ventricular repolarization

55-A normal QRS complex has a duration no greater than

- a) 0.1 seconds
- b) 0.2 seconds
- c) 0.3 seconds
- d) 0.4 seconds



Physiology: Dr. Ashabi, Dr. sohanaki, Dr. Rkhshan

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- b) aorta
- c) veins
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a

2- Which vessel has greatest blood velocity?

- a) Aorta
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- c) Arterioles
- d) Capillaries

c

velocity

Bf

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d

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- b) Blood pressure
- c) Blood flow
- d) Capacitance

c

DP